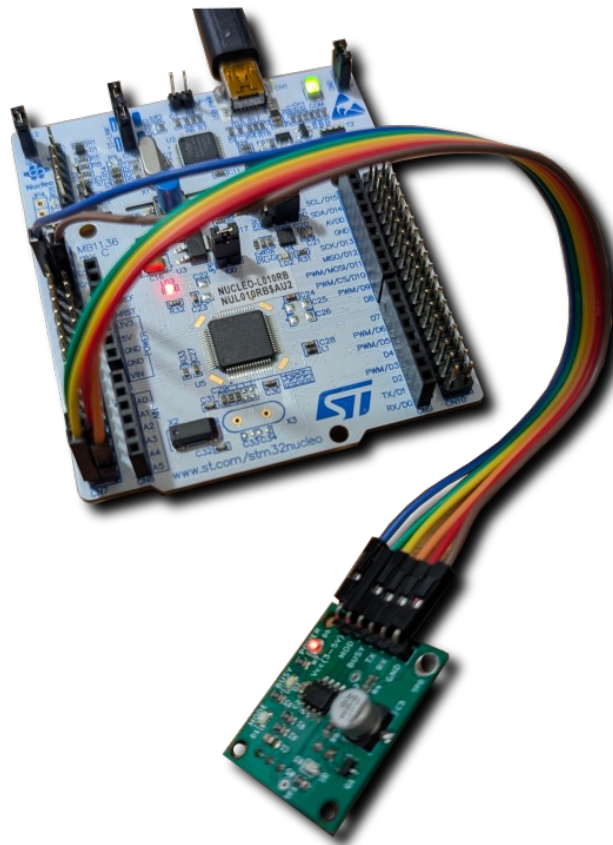


BC7215A AC Control Library

STM32 Demo Program

Manual



- Communication: Uses only serial ports (UART); compatible with any model in the STM32 series.
- Functionality: Demonstrates all library features, including air conditioner control and infrared (IR) command decoding.

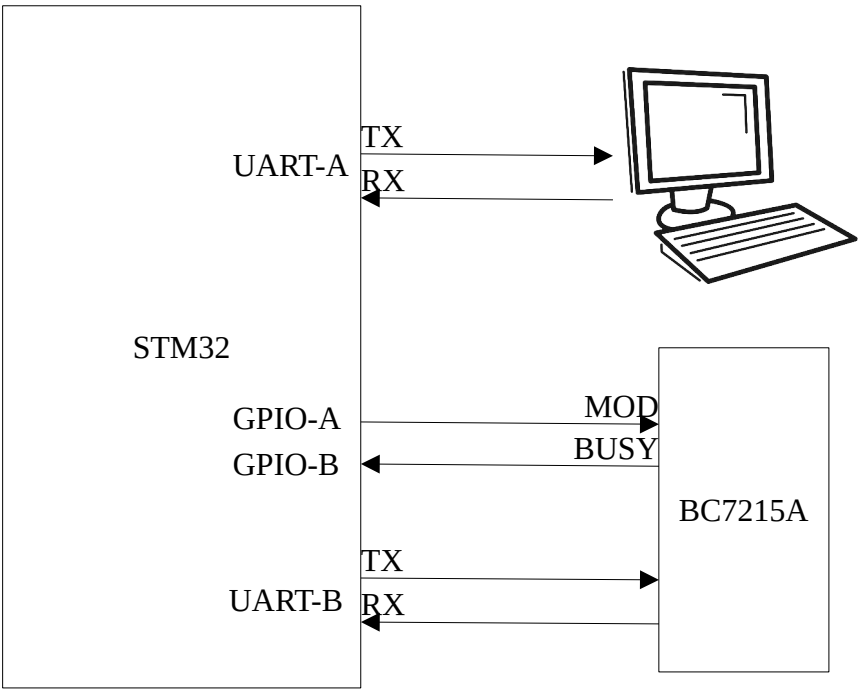
Introduction

This demo program is created based on STM32CubeIDE using the HAL library. In terms of hardware connection, it uses only two serial ports: one for the BC7215A and one for the PC. Users interact with the system via a serial terminal program or debugging tool. To port this to different MCU models, simply modify the .ioc file in STM32CubeIDE and update the corresponding configuration lines in main.c.

Hardware Connection and Configuration

Running the demo requires the following I/O resources:

- UART x2: Used to connect the BC7215A and the PC, respectively.
- GPIO x2: One configured as Input (connected to the BC7215A BUSY pin); one configured as Output (connected to the BC7215A MOD pin).



UART Port Parameter Configuration

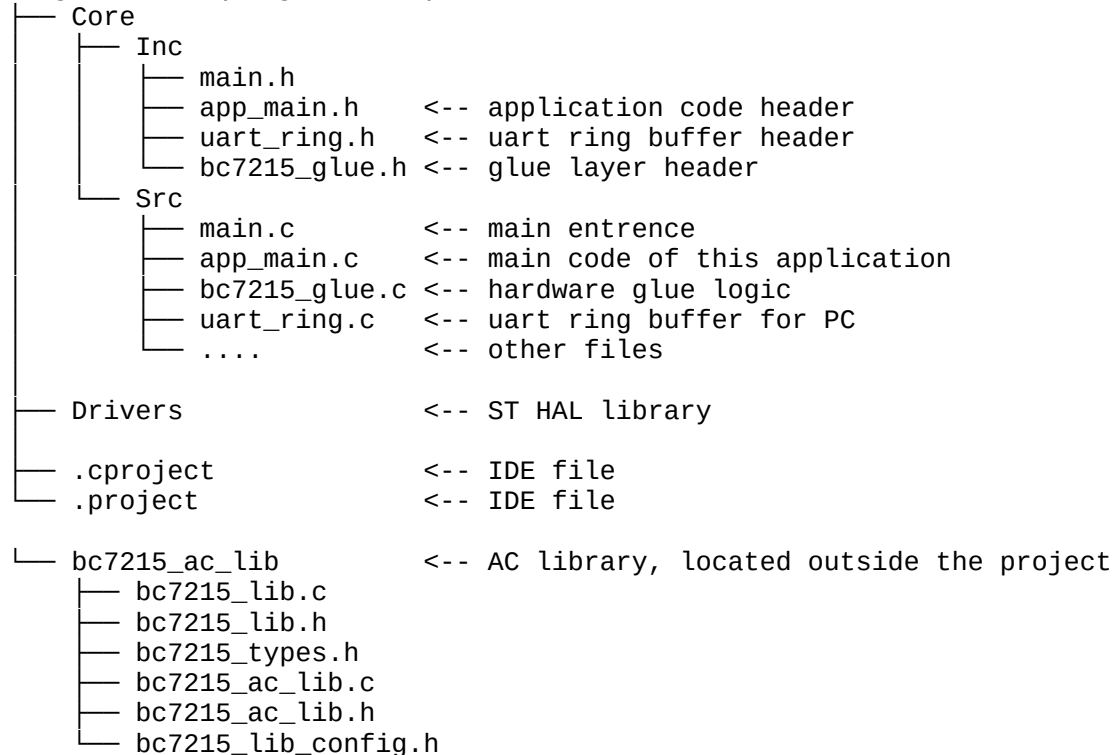
Connection Target	Configuration Settings
PC (Computer)	115200, 8N1, Asynchronous mode, Hardware Flow Control: Off, NVIC Interrupt: On, DMA: None.
BC7215A	19200, 8N2, Asynchronous mode, Hardware Flow Control: Off, NVIC Interrupt: On, DMA: None.

All other settings can remain at their default values. If in doubt, refer to main.c in the example code.

Directory Structure

The recommended structure in STM32CubeIDE is as follows:

Project_Name (Project root)

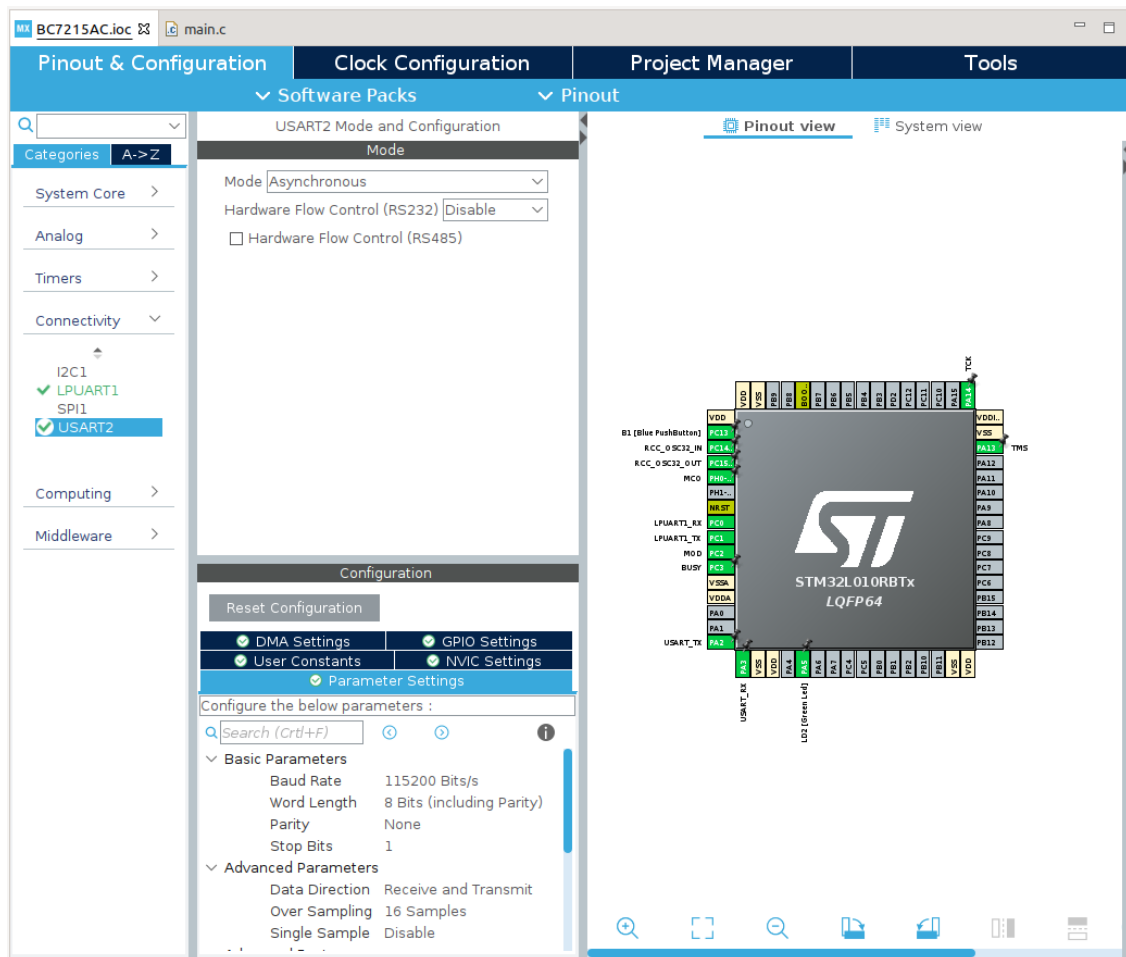


Reminder: The `bc7215_lib_config.h` file is tuned for the need of this STM32 example, check if it suits your application if you are using different system settings.

Porting to Different STM32 Processors

Due to the use of the HAL library, porting between STM32 series microprocessors is very simple and involves only two steps:

1. IDE Configuration: Select and set the UART and GPIO ports in the IDE based on your hardware, then synchronize these settings to generate code in main.c.



2. Code Update: In main.c, update the following section based on your hardware settings:

```

/* USER CODE BEGIN 2 */
//***** 移植时请根据
硬件环境修改此处设置
appConfig.consolePort = &huart2;
appConfig.bc7215Port = &hlpuart1;
appConfig.modPort = GPIOC;
appConfig.modPin = GPIO_PIN_2;
appConfig.bussyPort = GPIOC;
appConfig.bussyPin = GPIO_PIN_3;
App_Init(&appConfig);
/* USER CODE END 2 */

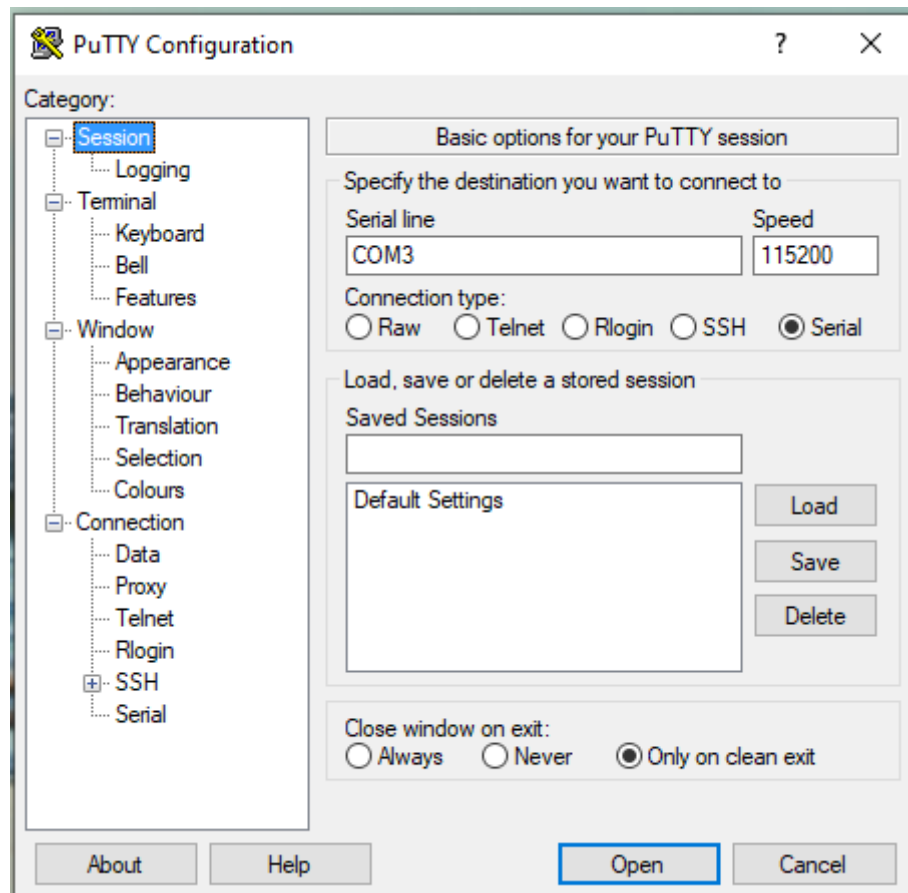
```

Software Usage

This example uses a serial connection to a PC for human-machine interaction via a terminal.

- Recommended Software: PuTTY for Windows or minicom for Linux.
- Alternative: General serial debugging tools can be used if they support ASCII mode and are set to automatically add CRLF (\r\n) when sending.
- Terminal Settings: 115200, 8N1, no flow control.

Note: If the terminal does not support UTF-8, the temperature "°C" symbol may appear garbled.



Instructions for Use

1. Initial Setup: Upon startup, the program will ask the user to choose between Celsius and Fahrenheit, as different functions are called for each system. If no display appears after opening the terminal, perform a hardware reset on the STM32.

File Edit View Terminal Tabs Help

System Initialized.

Before start, please choose your AC's temperature unit:

1. 0°C Celsius
2. 0°F Fahrenheit

2. Main Menu: The menu displays the library version and initialization status (whether it is paired with an AC unit).

File Edit View Terminal Tabs Help

* Welcome to BC7215A Universal AC Controller Demo *

AC Library Version: V5.4 build 2604

Current AC control library status: **!!! Not initialized !!!** (must be paired with AC before use)

Please select:

1. Pairing with AC
2. Control air conditioner
3. Save pairing data
4. Read saved data and pair with it
5. Try next match (if paired successfully but cannot control AC properly)
6. Load predefined protocol
7. Parse IR signal

1

Pairing: Please set AC remote to < Cooling mode, 250°C(770°F)>, and press Enter...

Now please aim at IR receiver and press <Fan Control> button on remote.

Will automatically proceed to next step after receiving signal...

Received 1 data packet(s):

Pkt 0 : 02 92 0F 00 00 00 F0 01 C2 FE 71 90 15 F0

Initialization (Pairing) **SUCCESS** !!!

Now press Enter to continue.

* Welcome to BC7215A Universal AC Controller Demo *

AC Library Version: V5.4 build 2604

Current AC control library status: ***** INITIALIZED *****

Please select:

1. Pairing with AC
2. Control air conditioner
3. Save pairing data
4. Read saved data and pair with it
5. Try next match (if paired successfully but cannot control AC properly)
6. Load predefined protocol
7. Parse IR signal

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7.1 | VT102 | Offline | ttyACM0

1. Pairing with AC

This must be completed before other functions can be used. The process merges IR signal capture and initialization.

2. Control Air Conditioner

Enter the sub-menu to change settings or toggle power.

```
File Edit View Terminal Tabs Help

*****
* Welcome to BC7215A Universal AC Controller Demo *
*****
AC Library Version: V5.4 build 2604
Current AC control library status: *** INITIALIZED ***
Please select:
  1. Pairing with AC
  2. Control air conditioner
  3. Save pairing data
  4. Read saved data and pair with it
  5. Try next match (if paired successfully but cannot control AC properly)
  6. Load predefined protocol
  7. Parse IR signal

2

AC Control, please select:
  1. Change A/C settings
  2. Power on
  3. Power off
  4. Return to upper menu

1

*** AC setting adjustment ***
Input format:
  Temperature, Mode, Fan, Pressed Key : (e.g. '24, 1, 2, 2')
Fewer parameter is allowed, for example '18, 2' means set to '18°C Heating',
Fan Speed and Pressed Key unchanged.
Temperature(°C)   Mode       Fan           Key
Range: 16~30     0 - Auto   0 - Auto     0 - Temp +
                  1 - Cool   1 - Low      1 - Temp -
                  2 - Heat   2 - Med      2 - Mode
                  3 - Dry    3 - High     3 - Fan
                  4 - Fan

* Values outside above ranges indicate maintaining current state for that item
-----
(Note: Limited to settings supported by the controlled AC.)
Now please enter AC parameter values: (enter 'exit' to return to upper menu)

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7.1 | VT102 | Offline | ttyACM0
```

In the control page, parameters are entered in the order: Temperature, Mode, Fan Speed, Pressed Key, separated by commas.

- Partial input is supported (e.g., entering only "18" will change only the temperature).
- Input "exit" to return to the upper menu.
- Values outside the legal range indicate that the current state for that item should be maintained.

```
File Edit View Terminal Tabs Help
*** AC setting adjustment ***
Input format:
  Temperature, Mode, Fan, Pressed Key : (e.g. '24, 1, 2, 2')
Fewer parameter is allowed, for example '18, 2' means set to '18°C Heating',
Fan Speed and Pressed Key unchanged.
Temperature(°C)   Mode       Fan       Key
Range: 16~30      0 - Auto   0 - Auto   0 - Temp +
                  1 - Cool   1 - Low    1 - Temp -
                  2 - Heat   2 - Med    2 - Mode
                  3 - Dry    3 - High   3 - Fan
                  4 - Fan
* Values outside above ranges indicate maintaining current state for that item
-----
(Note: Limited to settings supported by the controlled AC.)
Now please enter AC parameter values: (enter 'exit' to return to upper menu)

18
Sending command to set AC to: 18°C, Mode: Keep Fan: Keep, Key Pressed: Keep
Sending data: 02 92 0F 00 00 00 F0 01 D2 FE 71 20 15 F0
Transmission complete!
Input new value or 'exit' to return to upper level...
20, 2, 1
Sending command to set AC to: 20°C, Mode: Heat Fan: Low, Key Pressed: Keep
Sending data: 02 92 0F 00 00 00 F0 01 D2 FE 71 40 45 F0
Transmission complete!
Input new value or 'exit' to return to upper level...
19, 2, 3, 2
Sending command to set AC to: 19°C, Mode: Heat Fan: High, Key Pressed: Mode
Sending data: 02 92 0F 00 00 00 F0 01 B2 FE 71 30 4B F0
Transmission complete!
Input new value or 'exit' to return to upper level...
exit
Exit

AC Control, please select:
1. Change A/C settings
2. Power on
3. Power off
4. Return to upper menu

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7.1 | VT102 | Offline | ttyACM0
```

3/4. Save/Read Data

Because hardware varies, data storage methods are not universal; these functions are provided as placeholders for users to implement their own storage logic.

7. Parse IR Signal

Decodes "Temperature, Mode, Fan Speed, and Power Status" from an IR remote (only works for the remote paired with the system). Press Enter to return to the main menu.

File Edit View Terminal Tabs Help

1. Pairing with AC
2. Control air conditioner
3. Save pairing data
4. Read saved data and pair with it
5. Try next match (if paired successfully but cannot control AC properly)
6. Load predefined protocol
7. Parse IR signal

7

Receiving IR signal and parse(decode) Temperature, Mode, Fan Speed, and Power status from it.

BC7215A is now in RX mode, ready to decode. Press Enter exit

Parsing Result: Temp: 260°C, Mode: Cool, Fan Speed: Low, Power: On
Parsing Result: Temp: 270°C, Mode: Cool, Fan Speed: Low, Power: On
Parsing Result: Temp: 280°C, Mode: Cool, Fan Speed: Low, Power: On
Parsing Result: Temp: 200°C, Mode: Heat, Fan Speed: Auto, Power: On
Parsing Result: Temp: 250°C, Mode: Auto, Fan Speed: Auto, Power: On
Parsing Result: Temp: 220°C, Mode: Fan, Fan Speed: High, Power: On
Parsing Result: Temp: 230°C, Mode: Dry, Fan Speed: Low, Power: On
Parsing Result: Temp: 280°C, Mode: Cool, Fan Speed: Low, Power: On
Parsing Result: Temp: 280°C, Mode: Cool, Fan Speed: Low, Power: Off
Parsing Result: Temp: 280°C, Mode: Cool, Fan Speed: Low, Power: On

* Welcome to BC7215A Universal AC Controller Demo *

AC Library Version: V5.4 build 2604

Current AC control library status: *** INITIALIZED ***

Please select:

1. Pairing with AC
2. Control air conditioner
3. Save pairing data
4. Read saved data and pair with it
5. Try next match (if paired successfully but cannot control AC properly)
6. Load predefined protocol
7. Parse IR signal

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7.1 | VT102 | Offline | ttyACM0